SARR Pape Aldiouma

Cybersecurity Student, Year 3 | ISM Dakar

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PERSONAL DETAILS

Last name : SARR
First name : Pape Aldiouma
Date and place of birth : December 4, 2004, Camberène
Marital status : Single, no children
Address : Camberène, Dakar

ABOUT ME

Currently training in cybersecurity at ISM Dakar (Year 3), I put my skills into practice in network security, systems security, SOC/SIEM operations and incident response through supervised coursework and personal projects carried out in virtualized environments. Curious, rigorous and results-oriented, I am looking for an internship in cybersecurity to put these skills to work for an organization and keep growing in a demanding professional environment.

TECHNICAL SKILLS

- **Network and infrastructure security :** firewalling (OPNsense / pfSense), LAN/WAN/DMZ segmentation, IPsec VPN, NAT, packet filtering
- **Systems security :** Linux (Debian, Ubuntu) and Windows hardening, GPO, PAM/sudo, password policies
- **Container security :** Docker, isolation (namespaces, cgroups, seccomp), AppArmor, image scanning (Trivy), runtime detection (Falco)
- **SOC / SIEM and detection :** Elastic Stack (Kibana, ECS, KQL, EQL), detection engineering (Sigma), RBAC, dashboards, detection-as-code
- **Threat Intelligence and Threat Hunting :** MITRE ATT&CK, YARA, MISP, threat hunting
- **Digital forensics (DFIR) :** Volatility 3, Plaso, Auditd, chain of custody, NIST SP 800-61 methodology
- **Malware analysis :** static and dynamic analysis, isolated environment (INetSim), YARA rules
- **Ethical hacking and penetration testing :** Kali Linux (Nmap, Hydra, Nikto), lab-based testing
- **Virtualization :** QEMU/KVM, libvirt, virt-manager
- **Scripting :** Bash, PowerShell (basics)

CERTIFICATIONS

TryHackMe

Pre-Security — *Obtained in 2025*

Web Fundamentals — *Obtained in 2025*

Jr Penetration Tester — *Obtained in 2025*

Hack With Kali Linux

Hack With Kali Linux — *Obtained in 2025*

HackerX

Ethical Hacking & Cybersecurity — *Obtained in 2026*

Udemy

Ethical Hacking — *Obtained in 2026*

PROJECTS AND COURSEWORK

Personal projects (self-directed lab)

Advanced filtering and NAT configuration with OPNsense — *2025 to 2026*

Environment : *QEMU/KVM, OPNsense, Debian Client, Debian Server, Kali Linux*

- Created and managed an isolated virtual network under QEMU/KVM
- Set up advanced filtering rules on OPNsense (LAN, WAN, deny-all, aliases)
- Configured Outbound NAT (Hybrid mode) and Port Forward (port redirection)

- Simulated attacks from Kali Linux and validated the filtering rules
- Ran connectivity tests and analyzed firewall logs in real time

Deployment of Suricata IDS on OPNsense *2025 to 2026*

Environment : *QEMU/KVM, OPNsense, Debian Server, Kali Linux*

- Configured Suricata in PCAP live mode (IDS) on OPNsense
- Downloaded and enabled the ET Open and Abuse.ch rule sets
- Generated attack traffic from Kali Linux (Nmap, Hydra, Nikto, SQL injection)
- Read and interpreted alerts in real time (Alerts tab and EVE JSON logs)

Basic administration: user, permission and SSH management *2026*

Environment : *QEMU/KVM, Ubuntu Server, Kali Linux*

- Created and managed business accounts and groups, applying the principle of least privilege
- Set up fine-grained Unix permissions and ACLs on sensitive directories (finance, backup)
- Hardened SSH with ed25519 key pairs and configuration hardening (port, MaxAuthTries, AllowUsers)

Linux hardening: securing an Ubuntu server *2026*

Environment : *QEMU/KVM, Ubuntu Server, Kali Linux*

- Initial audit (active services, ports, Lynis score) and disabling of unnecessary services
- Configured the UFW firewall and anti-scan iptables rules for SSH
- Deployed Fail2Ban with custom jails
- Set up Auditd for traceability (identity files, sudoers, denied access)
- Hardened PAM/sudo (pam_pwquality, pam_faillock) and security sysctl parameters
- Compared Nmap scans before and after hardening

Protection against system attacks *2026*

Environment : *QEMU/KVM, Ubuntu Server, Kali Linux, Windows 11*

- Detected and quarantined malware with ClamAV, automated via Cron
- Mapped exposed services with Nmap and closed unnecessary ports
- Identified and remediated privilege escalation vectors (SUID, sudo NOPASSWD, cron)
- Performed forensic analysis of Auditd logs to trace malicious actions

Windows hardening: securing a workstation and a Windows server *2026*

Environment : *QEMU/KVM, Windows 11 Pro, Windows Server 2019, Kali Linux*

- Audited services, ports and security policies via PowerShell
- Disabled risky services (Print Spooler, Remote Registry, SMBv1, LLMNR, NetBIOS)
- Configured local GPOs (passwords, lockout, auditing) and the Windows Defender Firewall
- Deployed Sysmon with a custom configuration for advanced logging
- Hardened UAC and the registry (LM hash, administrative shares, anonymous access)

"Advanced Systems Security" module: SenTech Solutions lab (ISM Dakar)

Advanced isolation and containment *2026*

Tools : *Docker, AppArmor, seccomp, namespaces/cgroups, Trivy, Falco*

- Hardened Docker containers: reduced Linux capabilities, filtered system calls (seccomp), AppArmor confinement
- Scanned images for vulnerabilities with Trivy and detected runtime intrusions with Falco
- Simulated and then neutralized a container escape

Detection engineering and threat hunting *2026*

Tools : *MITRE ATT&CK, Sigma, YARA, MISP*

- Mapped attacker behaviors onto the MITRE ATT&CK framework
- Wrote Sigma detection rules (Linux and Windows) and YARA rules (file and memory)
- Conducted a threat hunt and captured the resulting intelligence as IOCs in MISP

Malware analysis *2026*

Tools : *YARA, MITRE ATT&CK, isolated lab (INetSim)*

- Built an isolated analysis lab with simulated network services (INetSim)
- Performed static analysis (strings, headers, introductory disassembly) and dynamic analysis (controlled, monitored execution) of samples
- Wrote an incident-response-style analysis report: IOCs, ATT&CK mapping, YARA detection rule

System forensics and incident response (DFIR) *2026*

Tools : *Volatility 3, Plaso, Auditd*

- Conducted a full DFIR investigation following NIST SP 800-61, from volatile evidence collection to a consolidated timeline
- Acquired and analyzed memory (Volatility 3), correlating it with disk artifacts
- Built a super-timeline (Plaso) and mapped the incident onto MITRE ATT&CK

"Ethical Hacking & SOC" module: SaaS track, Elastic Cloud Hosted (ISM Dakar)

Deploying a cloud SIEM *2026*

Tools : *Elastic Cloud Hosted*

- Deployed and managed an Elastic Cloud cluster (cost management, free trial, endpoint retrieval)

Data ingestion and search *2026*

Tools : *ECS template, Data View, KQL*

- Applied the ECS template and ingested simulated datasets
- Created a Data View, ran KQL searches and managed a restricted API key

Detection engineering & dashboards *2026*

Tools : *RBAC, EQL, detection-as-code*

- Managed spaces and RBAC, imported and tuned detection rules and dashboards
- Wrote custom detection rules and EQL queries, following a detection-as-code approach

SOC L1 incident investigation *2026*

Tools : *Triage, EQL correlation, Kibana Timeline, MITRE ATT&CK, Elastic Security Case*

- Triaged an alert and ran a full investigation (5W method), identifying IOCs and correlating events
- Reconstructed the incident timeline and mapped it onto MITRE ATT&CK
- Opened and documented an Elastic Security Case, writing a complete 13-section incident report

Project: OPNsense firewall, DMZ segmentation and roadwarrior IPsec VPN *2026*

- Designed and deployed a segmented network architecture (LAN / WAN / DMZ) under OPNsense
- Set up a roadwarrior IPsec VPN and firewall rules following the principle of least privilege
- Securely published a Docker application in the DMZ zone (Nginx reverse proxy)

PROFESSIONAL EXPERIENCE

Multiservice shop manager, Pikine Technologie (PIKTECH) *2023 to 2025*

3 consecutive seasons (July to October, each year)

- Welcomed and assisted customers with IT services
- Managed daily cash handling and equipment
- Provided first-level IT support and troubleshooting
- Built skills in customer relations and digital-space management

EDUCATION

Year 3, Cybersecurity, ISM Dakar *2026 to 2027 (ongoing)*

Year 2, Multimedia, Internet and Communication, ENO Pikine *2026 to 2027 (ongoing)*

Year 2, Cybersecurity, ISM Dakar *2025 to 2026*

Year 1, Multimedia, Internet and Communication, ENO Pikine *2025 to 2026*

Year 1, Cybersecurity, ISM Dakar *2024 to 2025*

High School Diploma (Baccalauréat), Groupe Scolaire Seydou Nourou Tall, Pikine *2023 to 2024*

BFEM, CEM Nioro Sud, Nioro *2019 to 2020*

CFEE, École Saint Jeanne Émilie de Villeneuve, Nioro (Kaolack) *2015 to 2016*

STRENGTHS

Curious and self-taught · Rigorous and organized · Team player · Resilient under pressure